Electronic Devices & Circuits Laboratory
Experiment 2: BJT Characteristics

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1 Aim

- To plot the input and output characteristics of a BJT in the Common Emitter (CE) configuration, and to determine the current gain β, input resistance r_i, and output resistance r_o.
- To plot the input and output characteristics of a BJT in the Common Base (CB) configuration, and to determine the current gain α, input resistance r_i, and output resistance r_o.
- To plot the input and output characteristics of a BJT in the Common Collector (CC) configuration, and to determine the current gain γ, input resistance r_i, and output resistance r_o.
- To verify the relationships $\beta = \alpha/(1-\alpha)$ and $\gamma = \beta + 1$ from the measured data.

2 Apparatus / Components

S.No.	Component / Equipment	Specification	Quantity
1	BC547 transistor	NPN, TO-92 package	1
2	Regulated DC power supply	0-30 V, dual output	2
3	Milliammeter	0-10 mA and 0-30 mA range	2
4	Microammeter	0-500 μA range	1
5	Voltmeter	0-1 V and 0-30 V range	2
6	Fixed resistors	1 kΩ, 4.7 kΩ, 10 kΩ	as needed
7	Breadboard, connecting wires, probes	Required	as needed

Table 1: List of components required.

3 Common Emitter (CE) Configuration

3.1 Theory

A Bipolar Junction Transistor (BJT) is a three-terminal device - emitter (E), base (B), and collector (C) - formed by two back-to-back PN junctions. In normal (active-region) operation, the base-emitter junction is forward-biased and the base-collector junction is reverse-biased. A small base current controls a much larger collector current, giving current gain.

In the CE configuration, the emitter is common to both input (base-emitter) and output (collector-emitter) loops. The input characteristic (I_B vs V_{BE} at constant V_{CE}) resembles a diode's forward characteristic. The output characteristic (I_C vs V_{CE} at constant I_B) shows a steep rise in the saturation region followed by a nearly flat active region, whose slight upward slope reflects a finite output resistance. The current gain $\beta = \Delta I_C/\Delta I_B$ is typically 50-300.

3.2 Circuit Connections and Procedure

Output Characteristics:

1. Connect the transistor in CE configuration: emitter grounded (common), base current supplied through a 10 kΩ resistor from the base supply with a microammeter (0-500 μA) in series, and collector connected through a milliammeter (0-10 mA) to the collector supply.

2. Connect a voltmeter (0-1 V) across base-emitter and a voltmeter (0-30 V) across collector-emitter.

3. Set the base supply to give $I_B = 20\mu A$ (adjust and confirm on the microammeter) and keep it fixed for this run.

4. Starting with $V_{CE} = 0$ V, increase V_{CE} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_C at each step.

5. Repeat for $I_B = 40, 60, 80$, and 100 μA, tabulating I_C vs V_{CE} for each. Switch off the supply.

Input Characteristics:

1. Retain the CE circuit. Set $V_{CE} = 1$ V and hold it fixed for this run.

2. Increase V_{BE} in steps of 0.05 V from 0 to 0.4 V, then in steps of 0.02 V up to 0.7 V, recording I_B at each step.

3. Repeat for $V_{CE} = 5$ V and $V_{CE} = 10$ V. Switch off the supply.

3.3 Circuit and Simulation

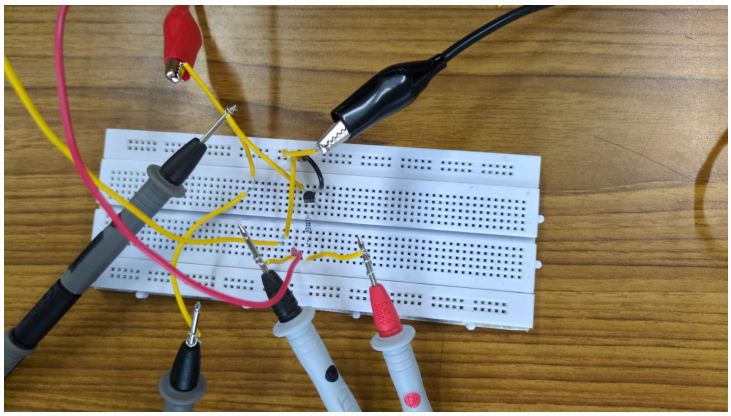


Figure 1: Hardware setup for CE configuration

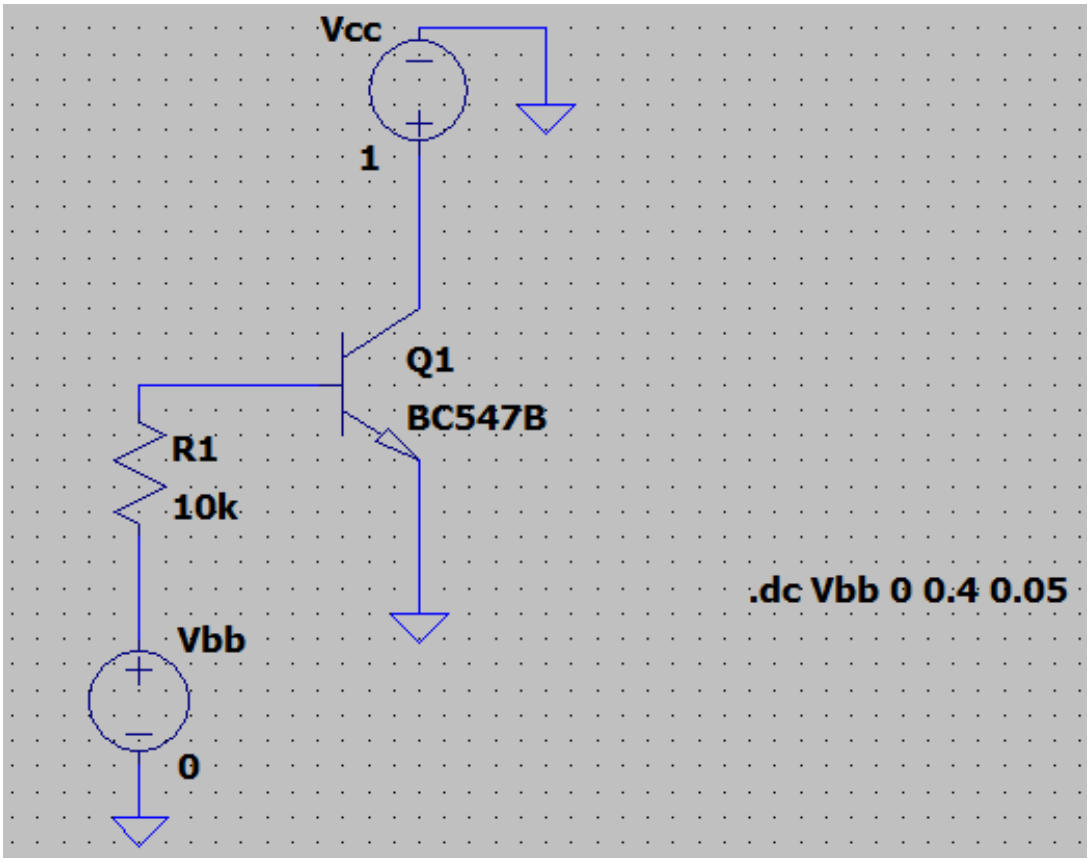


Figure 2: Simulation Circuit for CE Input Characteristics

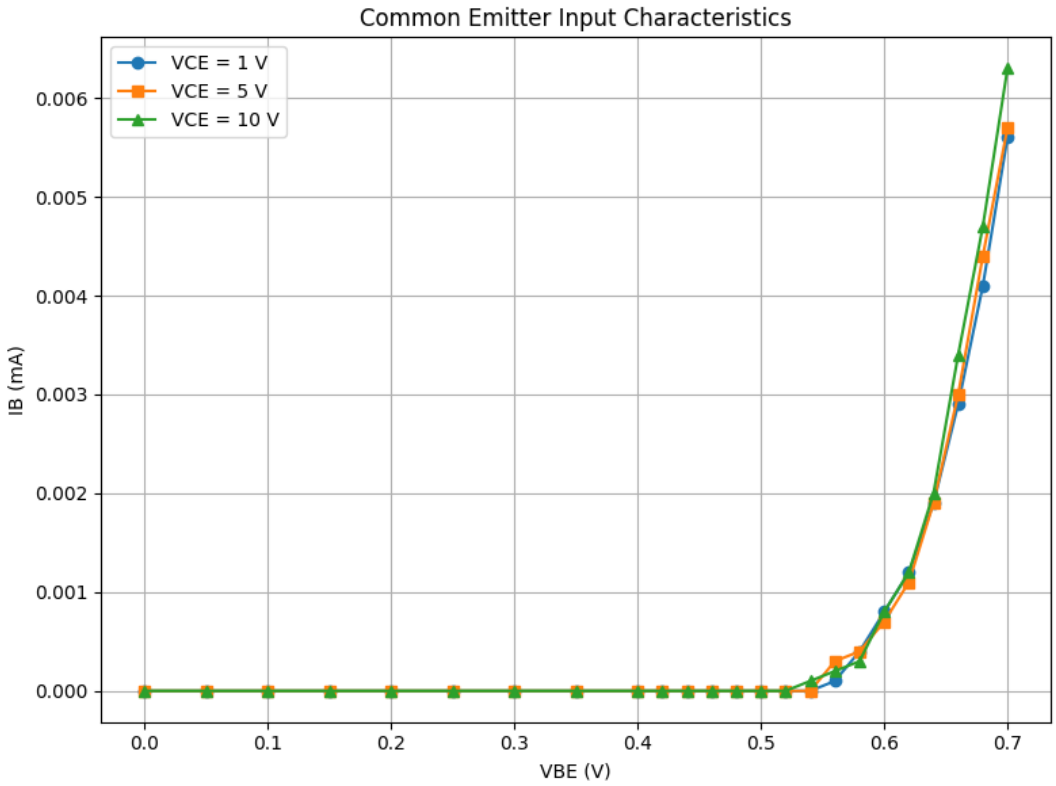


Figure 3: Simulated CE Input Characteristics (I_B vs V_{BE})

3.3.1 CE Input Simulations

The following simulations show the input characteristics at varying V_{CE} values.

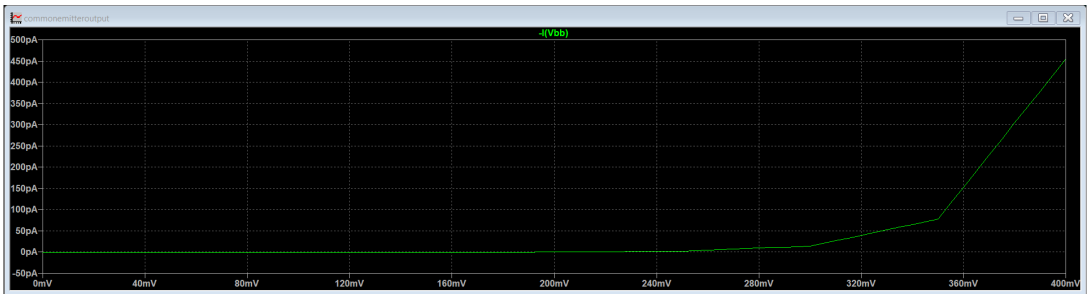


Figure 4: CE Input $V_{CE} = 1V$ (Low Range)

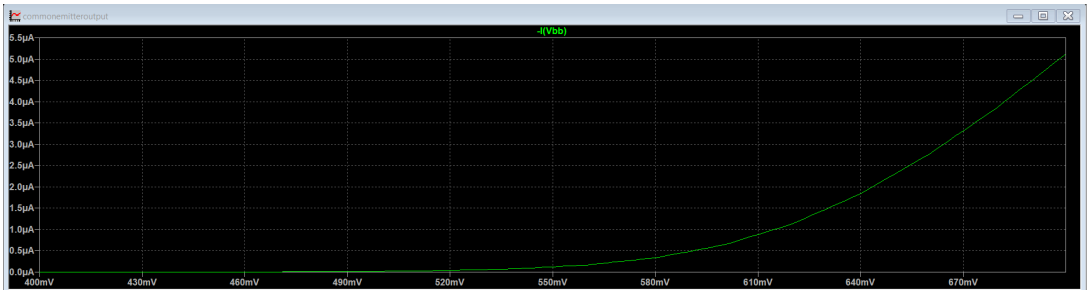


Figure 5: CE Input $V_{CE} = 1V$ (High Range)

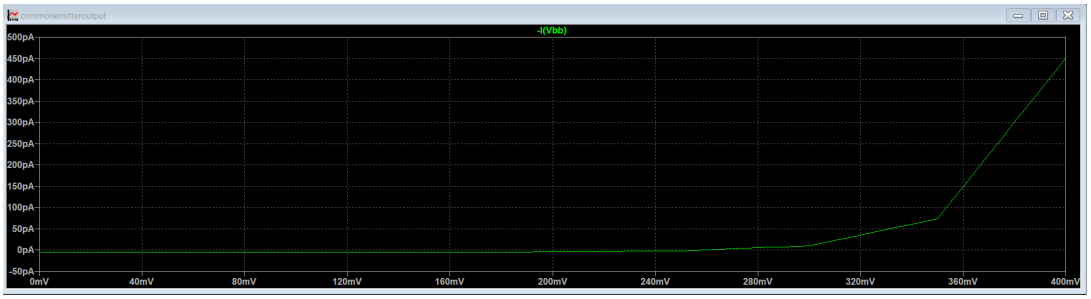


Figure 6: CE Input $V_{CE} = 5V$ (Low Range)

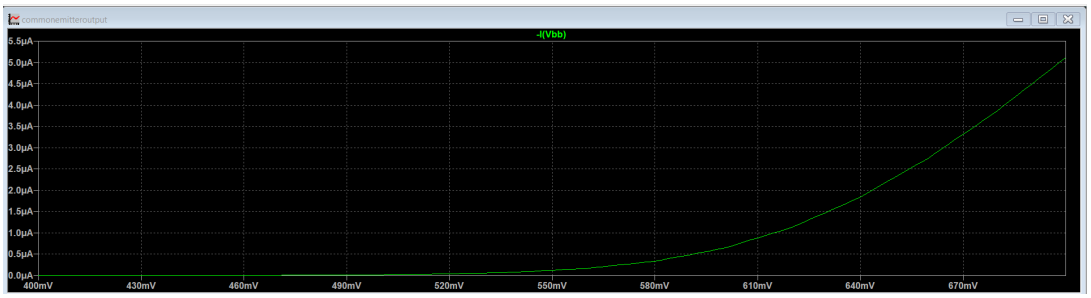


Figure 7: CE Input $V_{CE} = 5V$ (High Range)

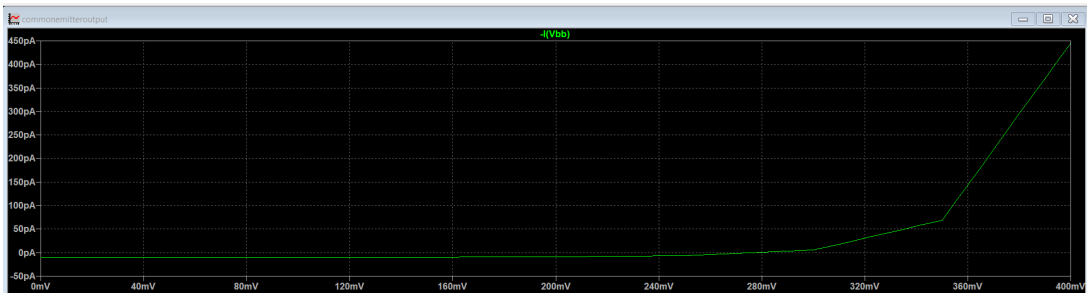


Figure 8: CE Input $V_{CE} = 10V$ (Low Range)

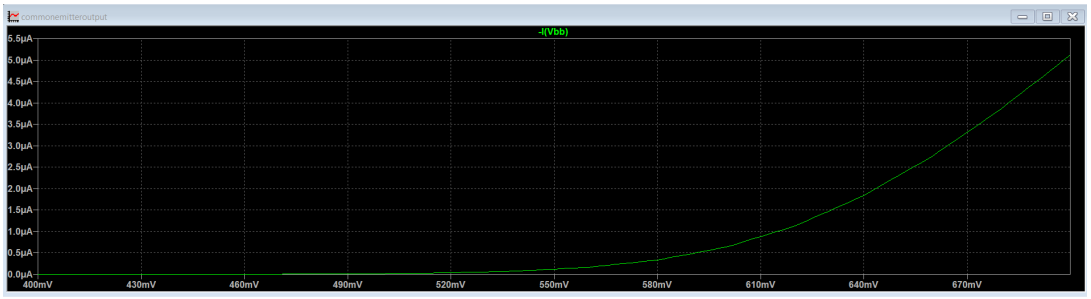


Figure 9: CE Input $V_{CE} = 10V$ (High Range)

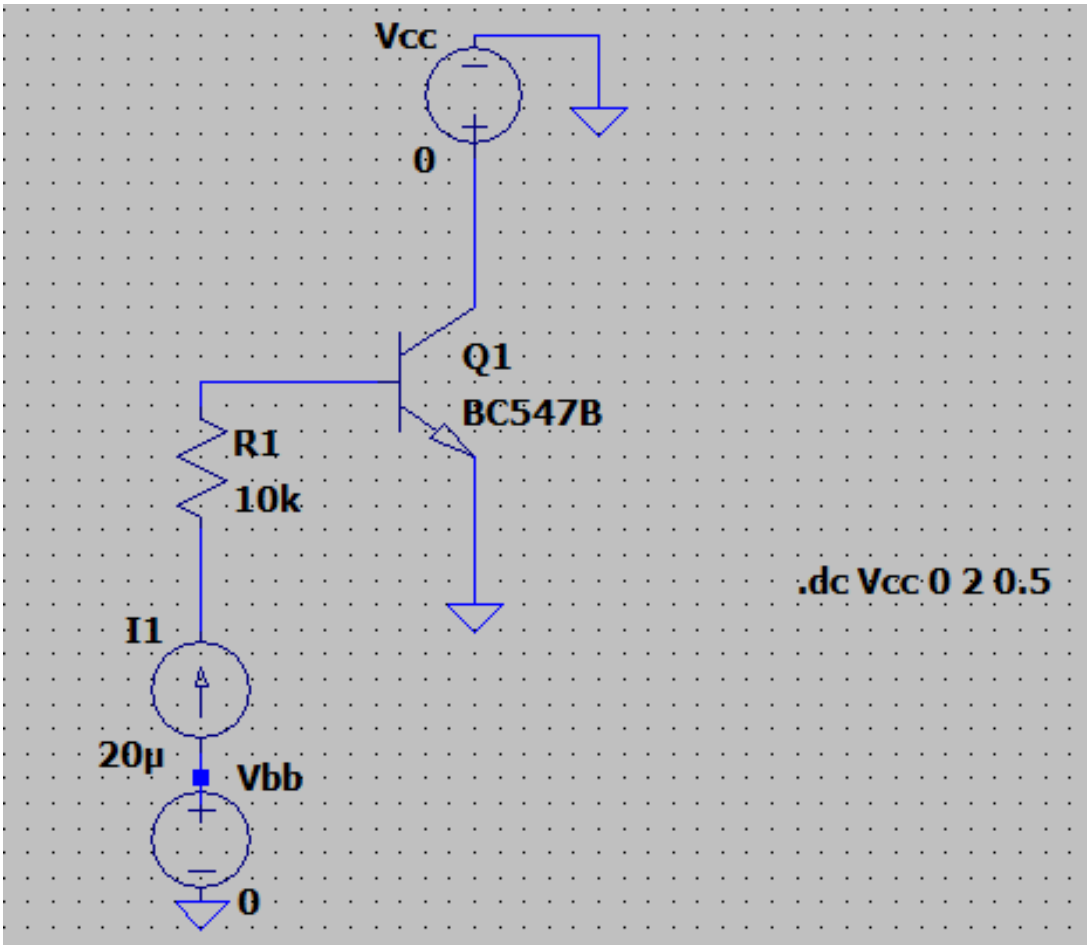


Figure 10: Simulation Circuit for CE Output Characteristics

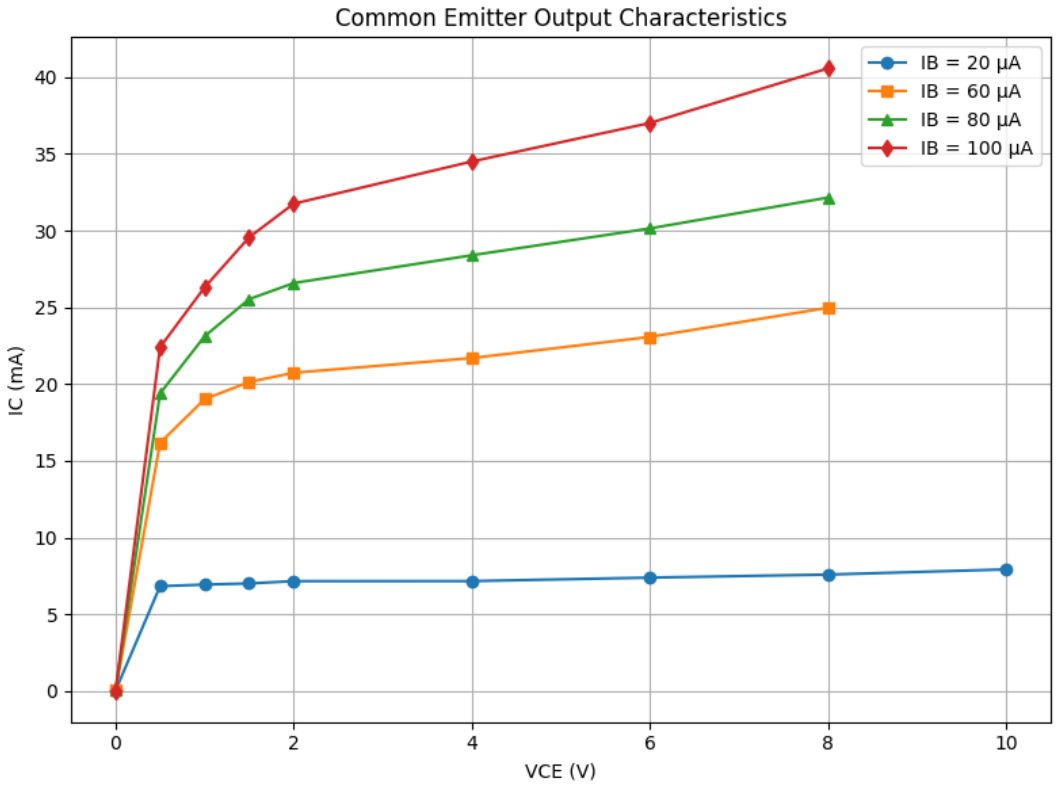


Figure 11: Simulated CE Output Characteristics (I_C vs V_{CE})

3.3.2 CE Output Simulations

The following images show the simulation results for varying base currents. The 'a' images represent the voltage sweep in steps of 0.5 V up to 2 V, and 'b' images represent steps of 2 V up to 20 V.

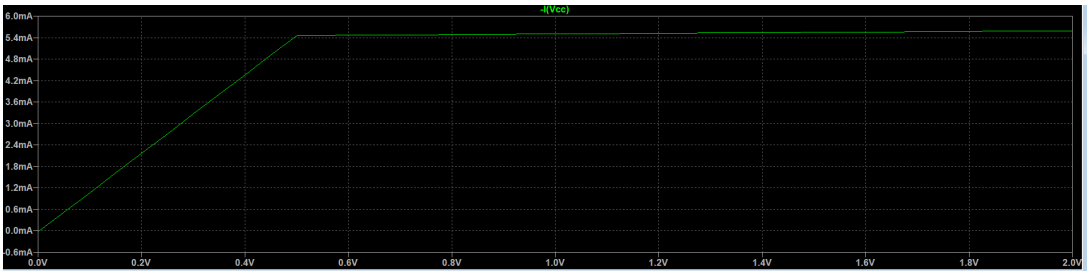


Figure 12: CE Output $I_B = 20\mu A$ (0 to 2V)

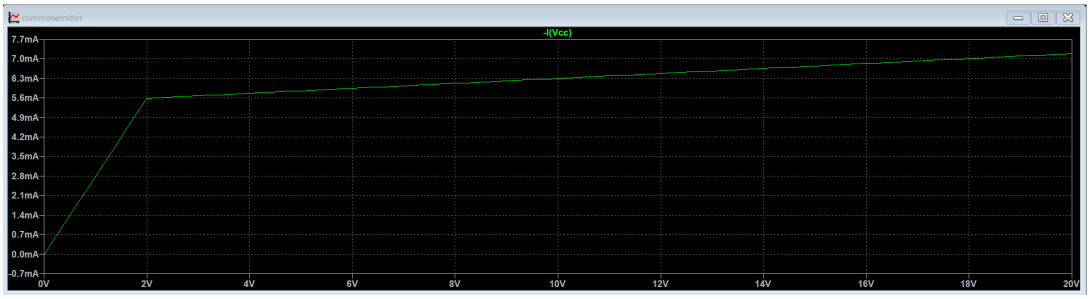


Figure 13: CE Output $I_B = 20\mu A$ (2 to 20V)

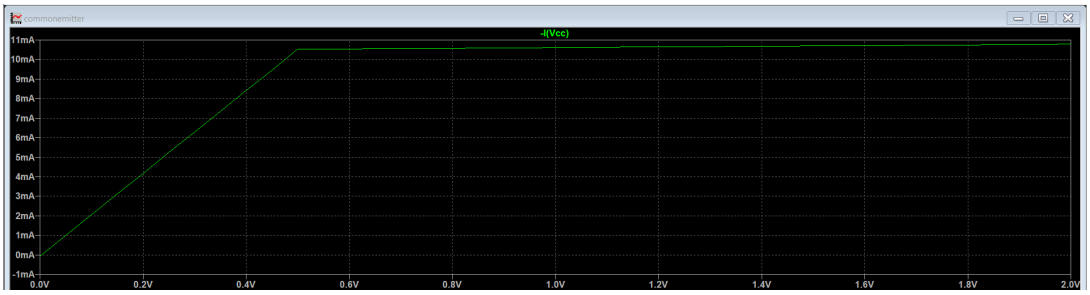


Figure 14: CE Output $I_B = 40\mu A$ (0 to 2V)

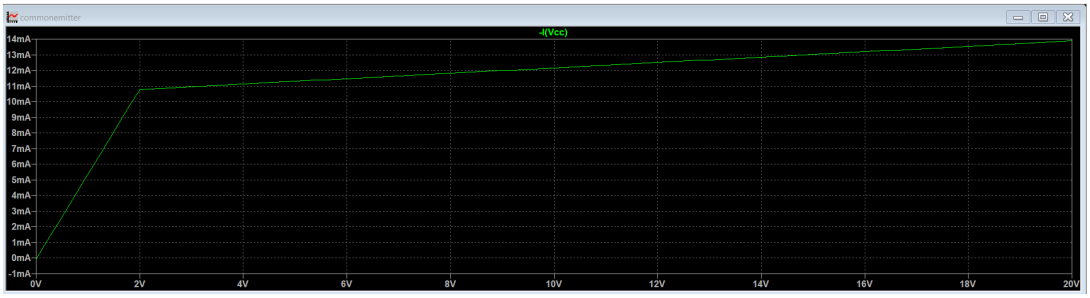


Figure 15: CE Output $I_B = 40\mu A$ (2 to 20V)

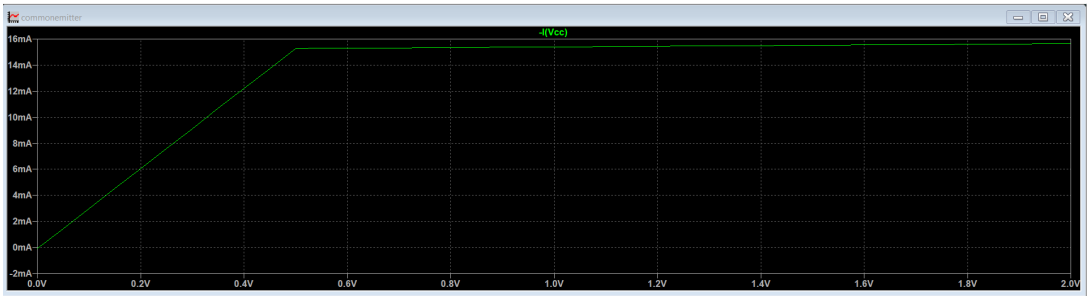


Figure 16: CE Output $I_B = 60\mu A$ (0 to 2V)

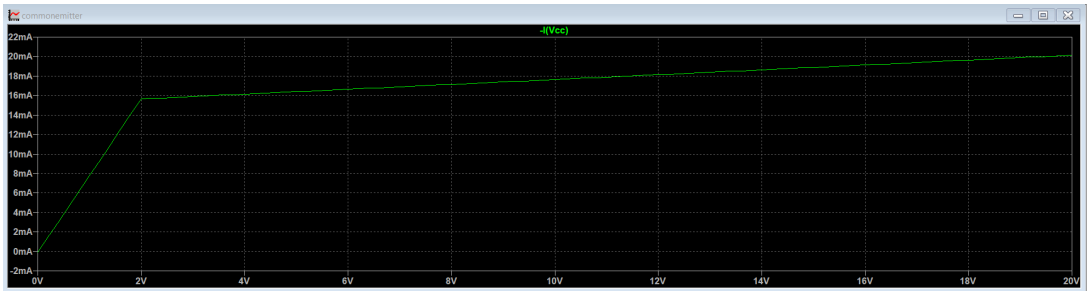


Figure 17: CE Output $I_B = 60\mu A$ (2 to 20V)

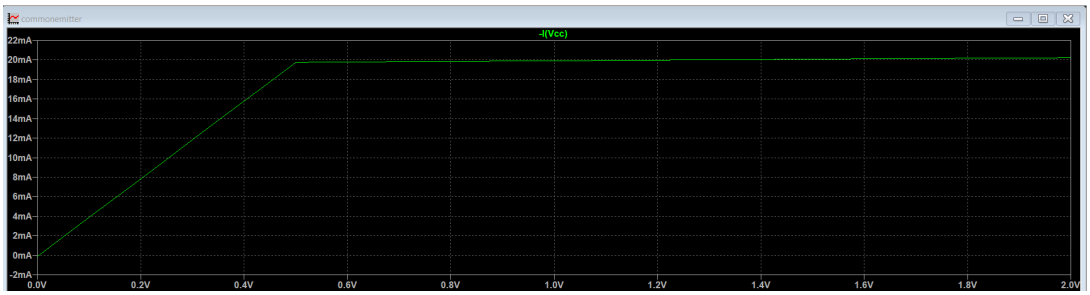


Figure 18: CE Output $I_B = 80\mu A$ (0 to 2V)

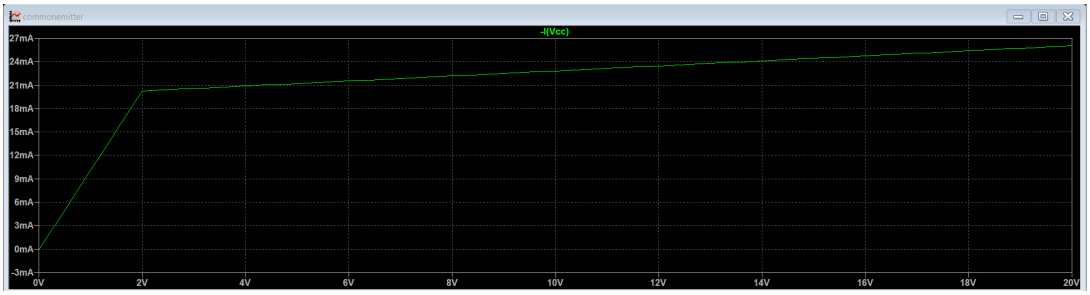


Figure 19: CE Output $I_B = 80\mu A$ (2 to 20V)

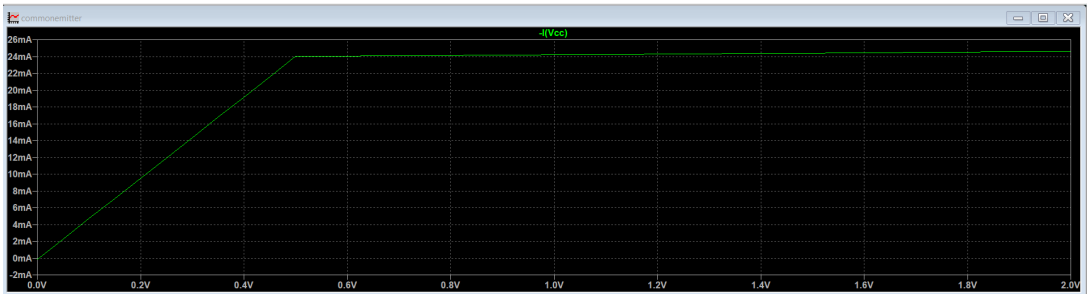


Figure 20: CE Output $I_B = 100\mu A$ (0 to 2V)

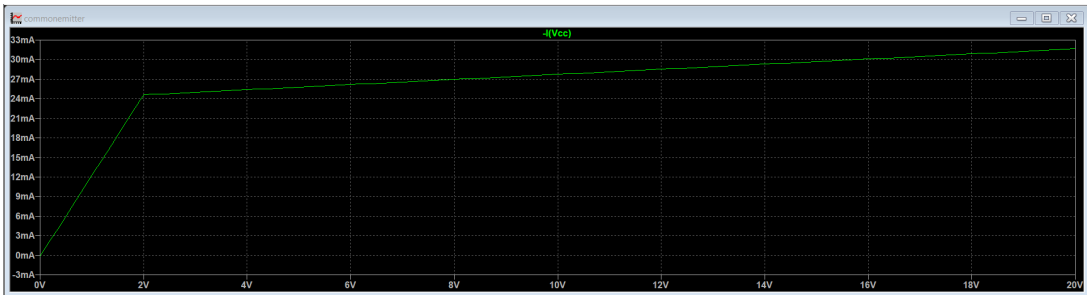


Figure 21: CE Output $I_B = 100\mu A$ (2 to 20V)

3.4 Observation Tables

Table 2: CE Output Characteristics

S.No.	V_{CE} **(V)**	I_C **(mA) at** $I_B = 20\mu A$	I_C **(mA) at** $I_B = 60\mu A$	I_C **(mA) at** $I_B = 100\mu A$
1	0.0	0.031	0.031	0.0026
2	0.5	6.825	16.170	22.4000
3	1.0	6.930	19.015	26.3010
4	1.5	7.005	20.132	29.5600
5	2.0	7.155	20.732	31.7500
6	4.0	7.158	21.691	34.5000
7	6.0	7.385	23.070	37.0100
8	8.0	7.583	24.970	40.5700

Table 3: CE Input Characteristics

S.No.	V_{BE} **(V)**	I_B **(**μ**A) at** $V_{CE} = 1V$	I_B **(**μ**A) at** $V_{CE} = 5V$	I_B **(**μ**A) at** $V_{CE} = 10V$
1	0.00	0.0	0.0	0.0
2	0.05	0.0	0.0	0.0
3	0.10	0.0	0.0	0.0
4	0.15	0.0	0.0	0.0
5	0.20	0.0	0.0	0.0
6	0.25	0.0	0.0	0.0
7	0.30	0.0	0.0	0.0
8	0.35	0.0	0.0	0.0

4 Common Base (CB) Configuration

4.1 Theory

In the CB configuration, the base is common to both loops. The input characteristic (I_E vs V_{EB} at constant V_{CB}) is again diode-like. The output characteristic (I_C vs V_{CB} at constant I_E) is extremely flat, since the collector current is set almost entirely by the emitter current and is nearly independent of V_{CB} giving CB circuits a very high output resistance. The current gain $\alpha = \Delta I_C / \Delta I_E$ is always slightly less than 1 (typically 0.95-0.99).

4.2 Circuit Connections and Procedure

Output Characteristics:

1. Connect the transistor in CB configuration: base grounded (common), emitter current supplied through a 1 kΩ resistor from the emitter supply with a milliammeter (0-10 mA) in series, and collector connected through a milliammeter (0-10 mA) to the collector supply.

2. Connect a voltmeter (0-1 V) across emitter-base and a voltmeter (0-30 V) across collector-base.

3. Set the emitter supply to give $I_E = 2$ mA and hold it fixed for this run.

4. Starting with $V_{CB} = 0$ V, increase V_{CB} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_C at each step.

5. Repeat for $I_E = 4, 6, 8$, and 10 mA. Switch off the supply.

Input Characteristics:

1. Retain the CB circuit. Set $V_{CB} = 0$ V and hold it fixed for this run.

2. Increase V_{EB} in steps of 0.05 V from 0 to 0.4 V, then in steps of 0.02 V up to 0.7 V, recording I_E at each step.

3. Repeat for $V_{CB} = 5$ V and $V_{CB} = 10$ V. Switch off the supply.

4.3 Circuit and Simulation

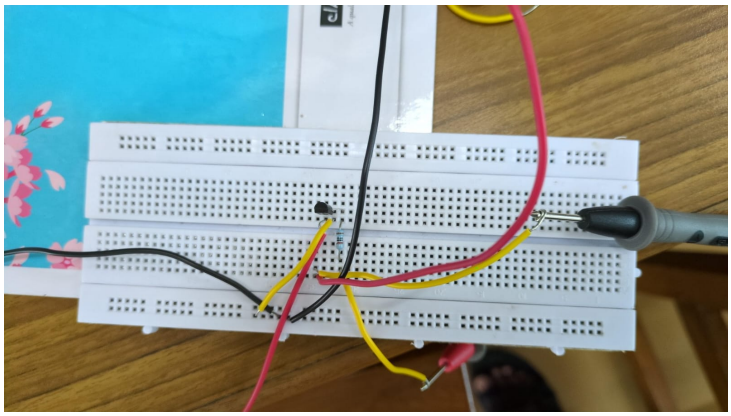


Figure 22: Hardware setup for CB configuration

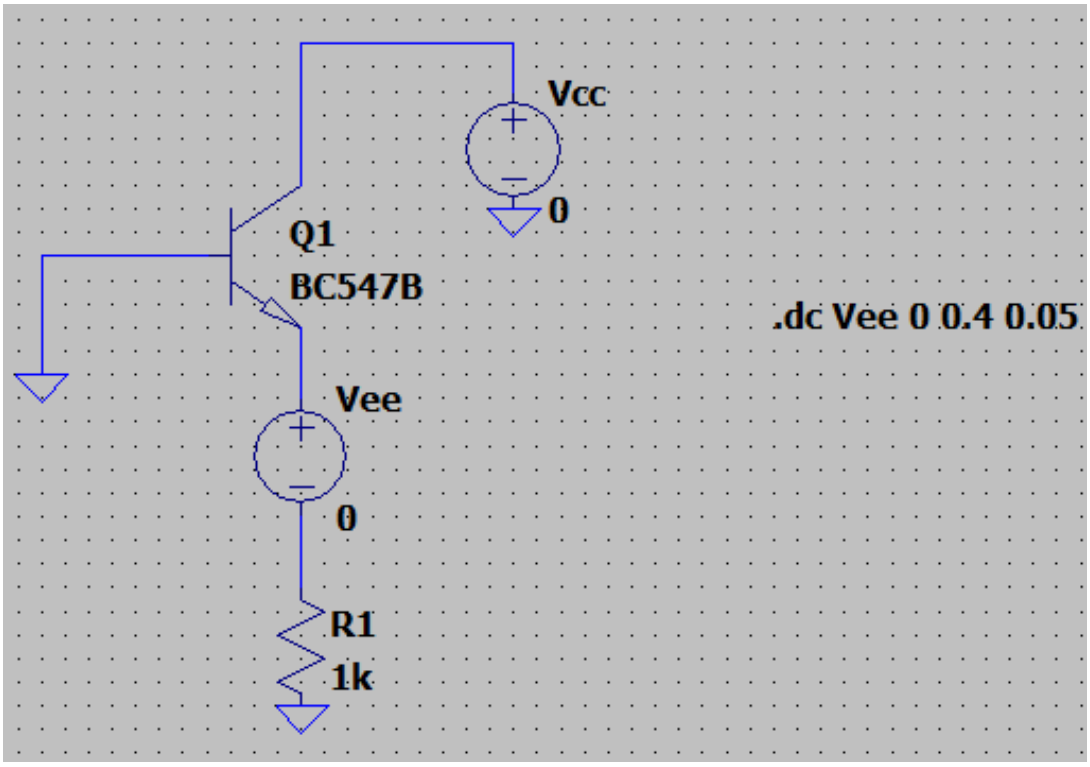


Figure 23: Simulation Circuit for CB Configuration

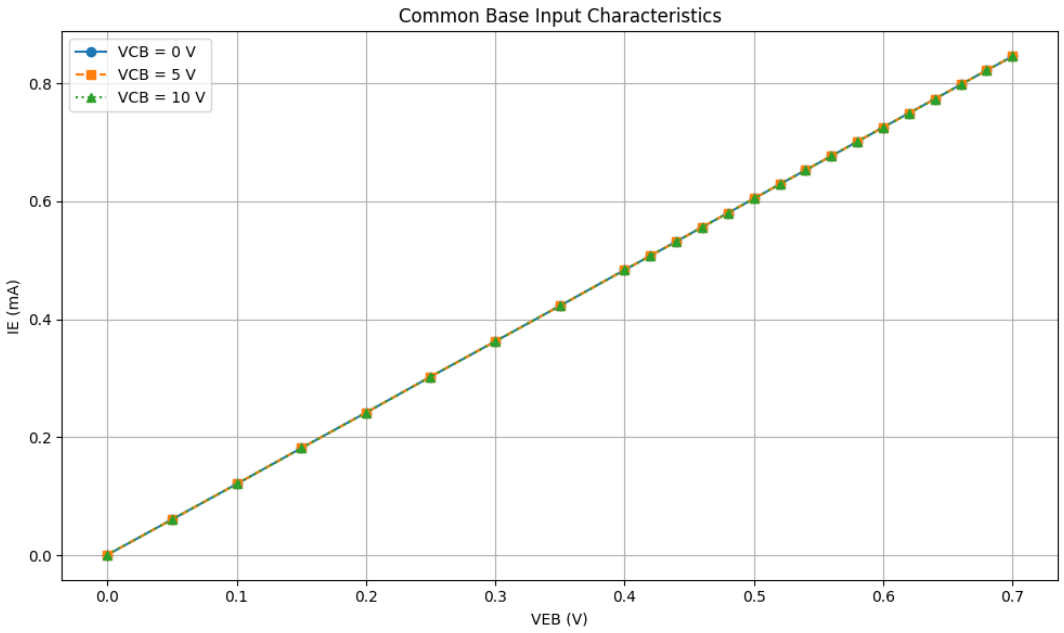


Figure 24: Simulated CB Input Characteristics (I_E vs V_{EB})

4.3.1 CB Input Simulations

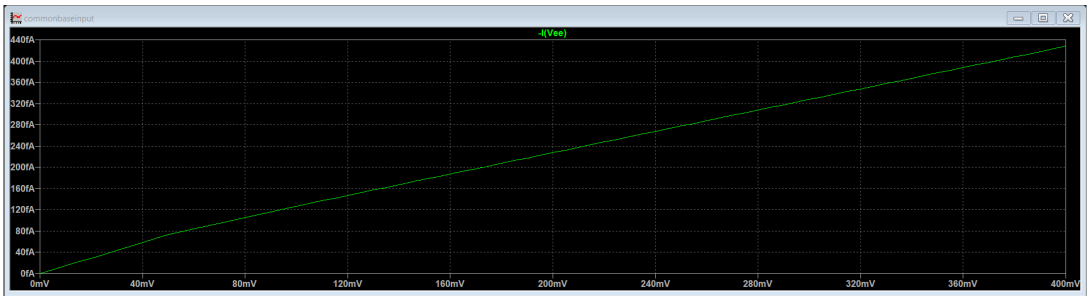


Figure 25: CB Input $V_{CB} = 0V$ (Low Range)

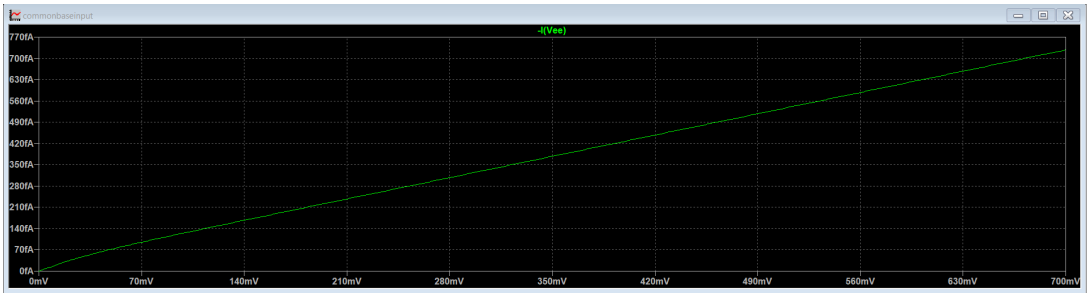


Figure 26: CB Input $V_{CB} = 0V$ (High Range)

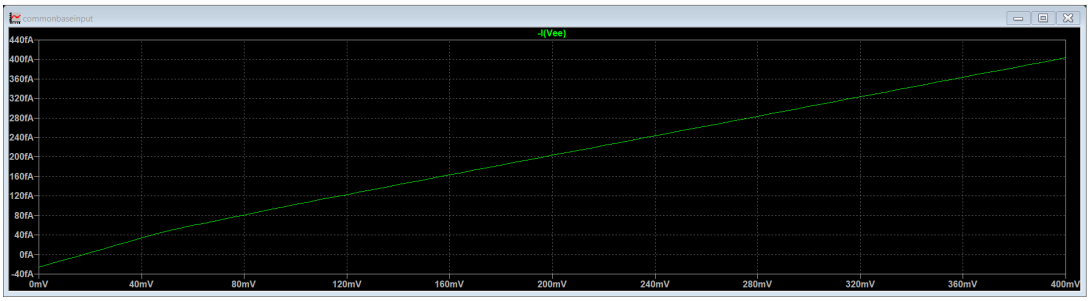


Figure 27: CB Input $V_{CB} = 5V$ (Low Range)

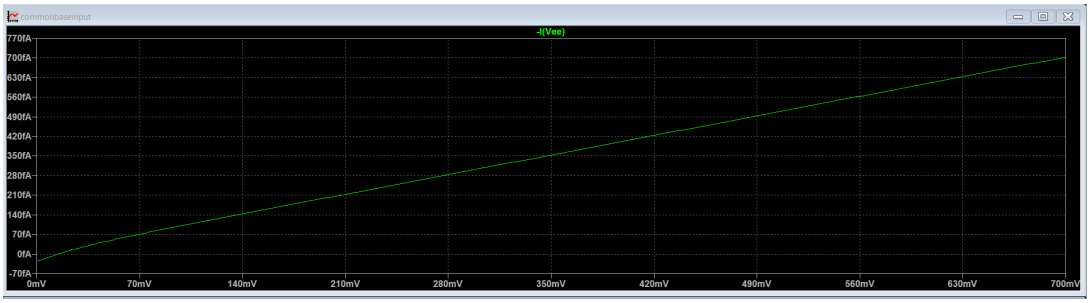


Figure 28: CB Input $V_{CB} = 5V$ (High Range)

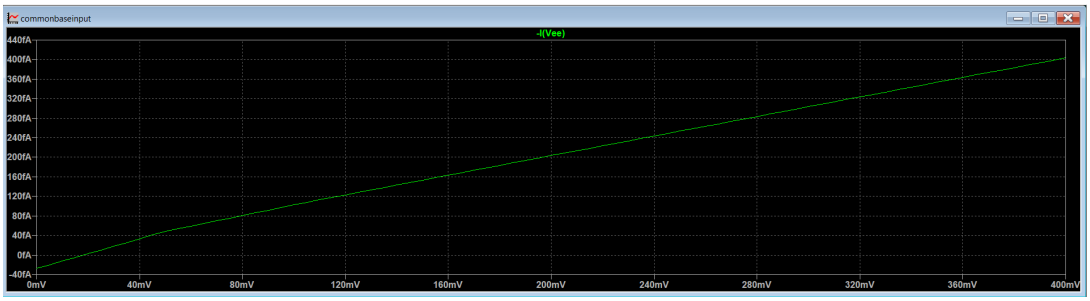


Figure 29: CB Input $V_{CB} = 10V$ (Low Range)

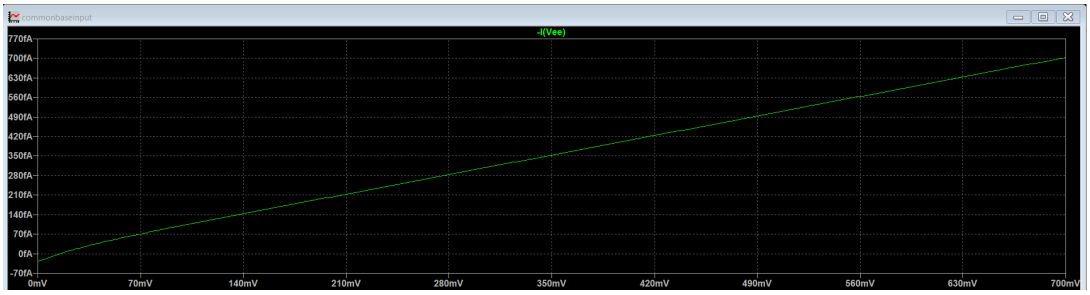


Figure 30: CB Input $V_{CB} = 10V$ (High Range)

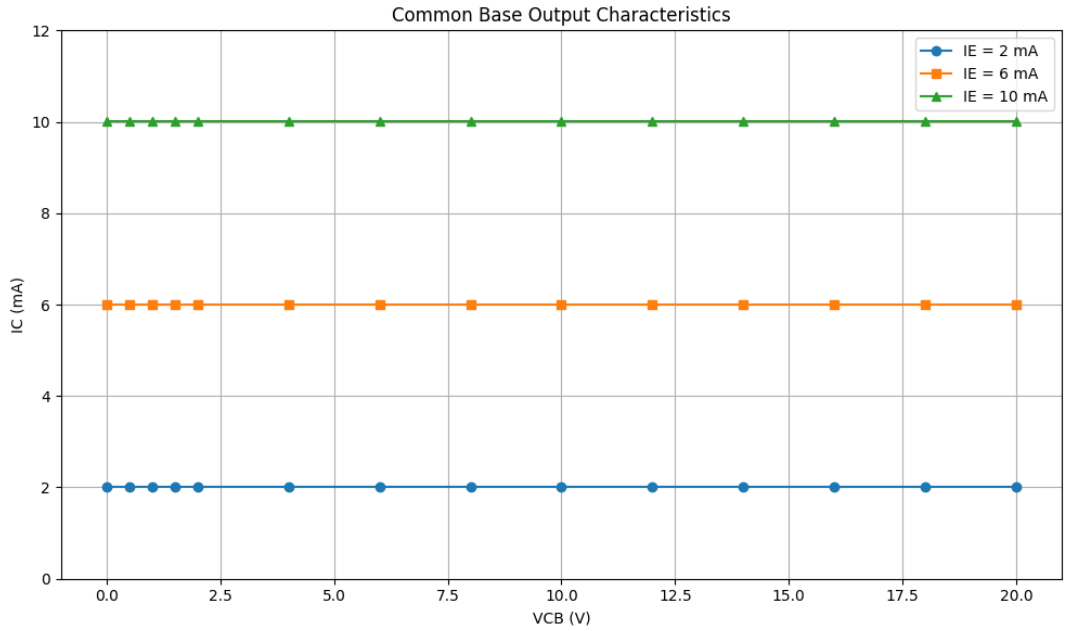


Figure 31: Simulated CB Output Characteristics (I_C vs V_{CB})

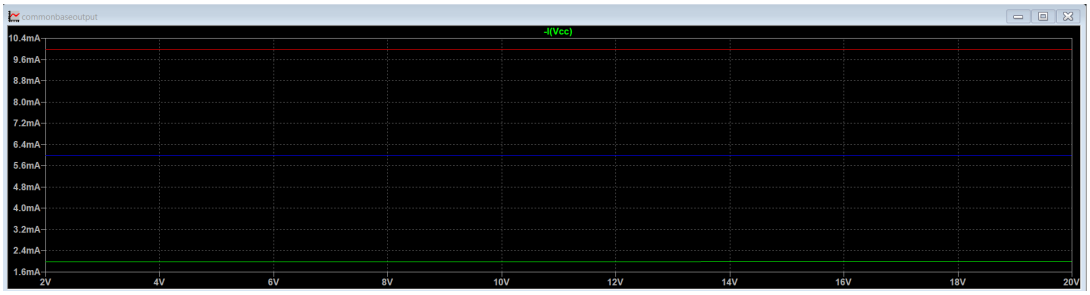


Figure 32: Simulated CB Output Characteristics Detailed Simulation

4.4 Observation Tables

Table 4: CB Output Characteristics

S.No.	V_{CB} **(V)**	I_C **(mA) at** $I_E = 2$ **mA**	I_C **(mA) at** $I_E = 6$ **mA**	I_C **(mA) at** $I_E = 10$ **mA**
1	0.0	2.01	6.01	10.01
2	0.5	2.01	6.01	10.01
3	1.0	2.01	6.01	10.01
4	1.5	2.01	6.01	10.01
5	2.0	2.01	6.01	10.01
6	4.0	2.01	6.01	10.01
7	6.0	2.01	6.01	10.01
8	8.0	2.01	6.01	10.01

Table 5: CB Input Characteristics

S.No.	V_{EB} **(V)**	I_E **(mA) at** $V_{CB} = 0V$	I_E **(mA) at** $V_{CB} = 5V$	I_E **(mA) at** $V_{CB} = 10V$
1	0.00	0.0007	0.0008	0.0008
2	0.05	0.0606	0.0606	0.0606
3	0.10	0.1207	0.1207	0.1207
4	0.15	0.1816	0.1816	0.1816
5	0.20	0.2417	0.2417	0.2417
6	0.25	0.3026	0.3026	0.3026
7	0.30	0.3625	0.3625	0.3625
8	0.35	0.4227	0.4227	0.4227

5 Common Collector (CC) Configuration

5.1 Theory

In the CC configuration (also called an emitter follower), the collector is common to both loops. The output characteristic (I_E vs V_{EC} at constant I_B) closely resembles the CE output characteristic, since $I_E \approx I_C$. The input characteristic is normally plotted as I_B vs V_{BC} at constant V_{EC}, and falls off as the base-collector reverse bias increases. The current gain $\gamma = \Delta I_E / \Delta I_B \approx \beta + 1$, so CC circuits provide current gain close to that of the CE configuration but are used mainly for their high input resistance and low output resistance (voltage buffering).

5.2 Circuit Connections and Procedure

Output Characteristics:

1. Connect the transistor in CC configuration: collector grounded (common), base current supplied through a 10 kΩ resistor from the base supply with a microammeter (0-500 μA) in series, and emitter connected through a milliammeter (0-10 mA) to the emitter supply.

2. Connect a voltmeter (0-30 V) across base-collector and a voltmeter (0-30 V) across emitter-collector.

3. Set the base supply to give $I_B = 20\mu A$ and hold it fixed for this run.

4. Starting with $V_{EC} = 0$ V increase V_{EC} in steps of 0.5 V up to 2 V, then in steps of 2 V up to 20 V, recording I_E at each step.

5. Repeat for $I_B = 40, 60, 80$, and 100 μA. Switch off the supply.

Input Characteristics:

1. Retain the CC circuit. Set $V_{EC} = 1$ V and hold it fixed for this run.

2. Increase V_{BC} in steps of 0.5 V from 0 to 10 V, recording I_B at each step.

3. Repeat for $V_{EC} = 5$ V and $V_{EC} = 10$ V. Switch off the supply.

5.3 Circuit and Simulation

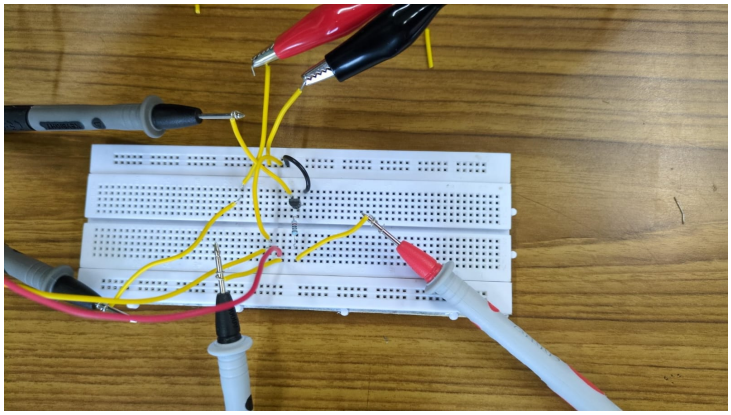


Figure 33: Hardware setup for CC configuration

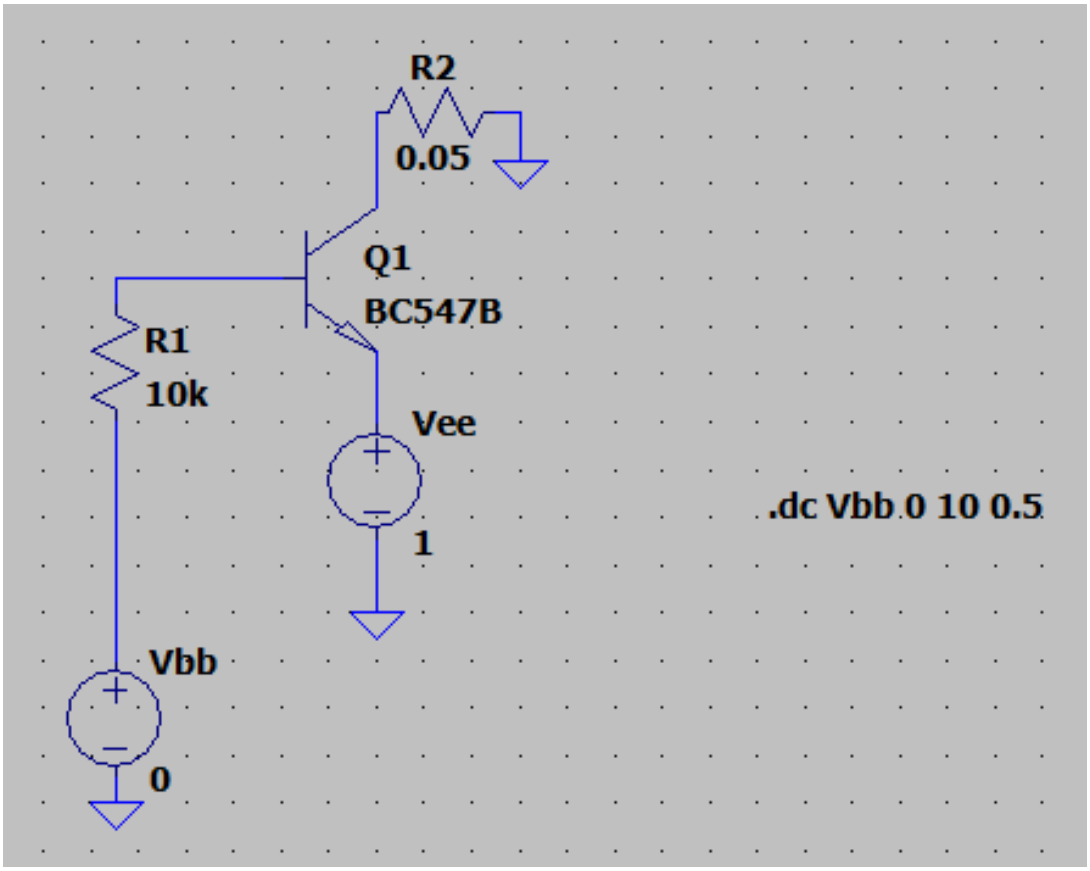


Figure 34: Simulation Circuit for CC Configuration

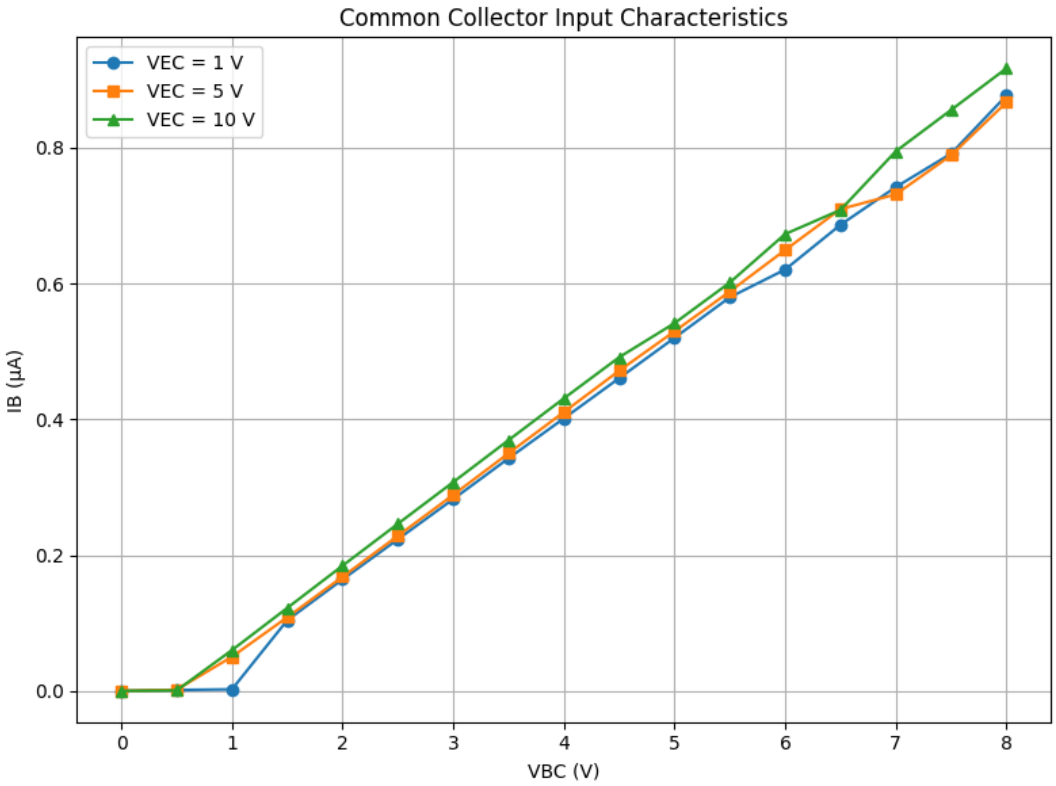


Figure 35: Simulated CC Input Characteristics (I_B vs V_{BC})

5.3.1 CC Input Simulations

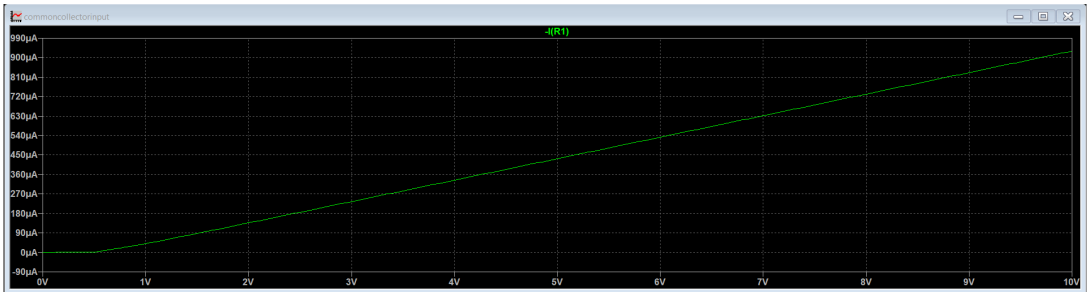


Figure 36: CC Input $V_{EC} = 1V$

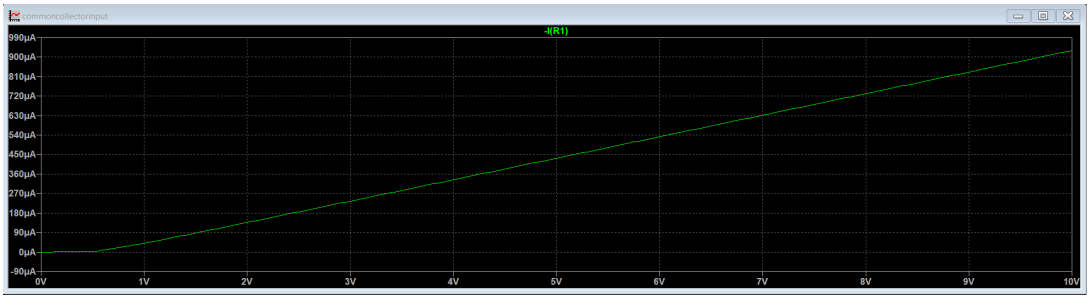


Figure 37: CC Input $V_{EC} = 5V$

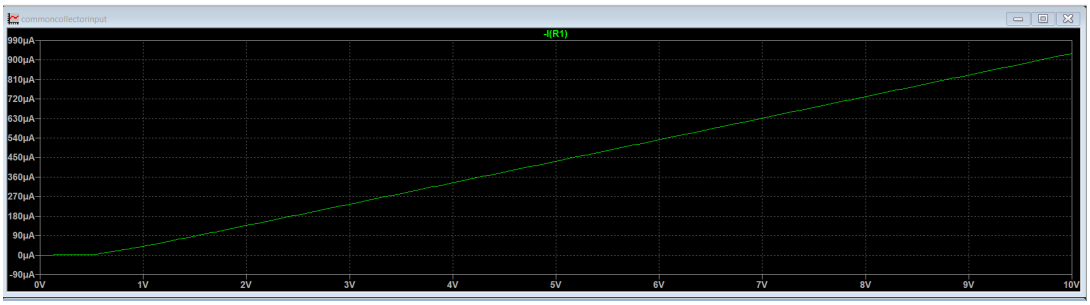


Figure 38: CC Input $V_{EC} = 10V$

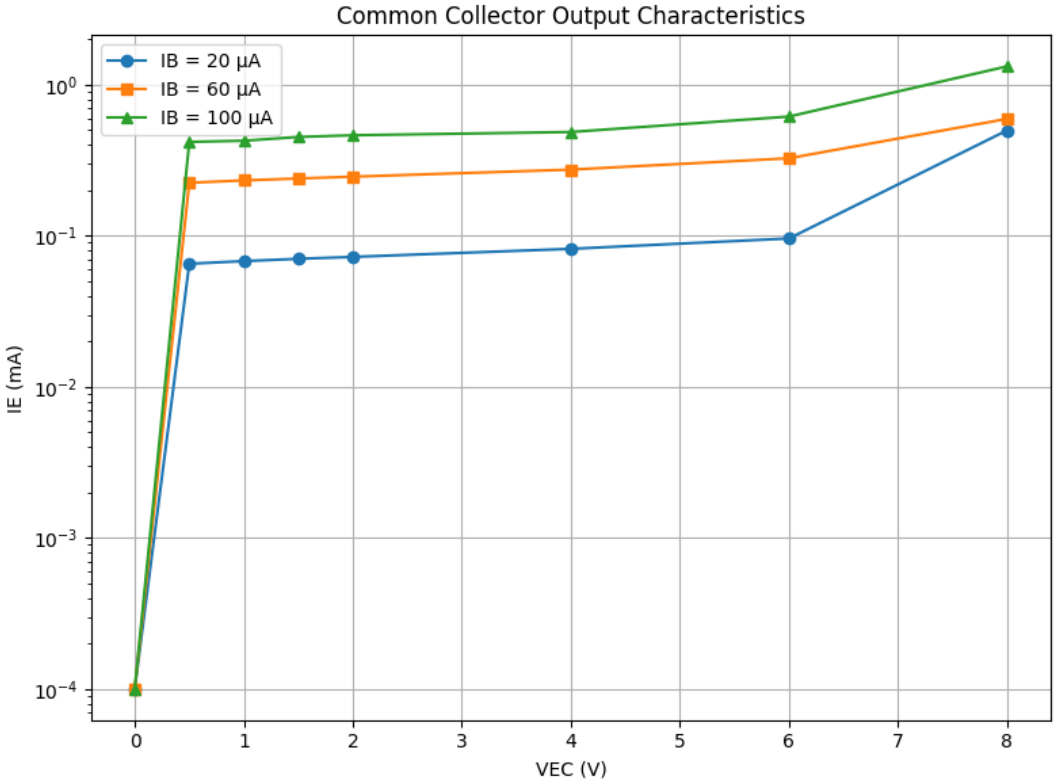


Figure 39: Simulated CC Output Characteristics (I_E vs V_{EC})

5.3.2 CC Output Simulations

The following images show the simulation results for varying base currents. The 'a' images represent the voltage sweep in steps of 0.5 V up to 2 V, and 'b' images represent steps of 2 V up to 20 V.

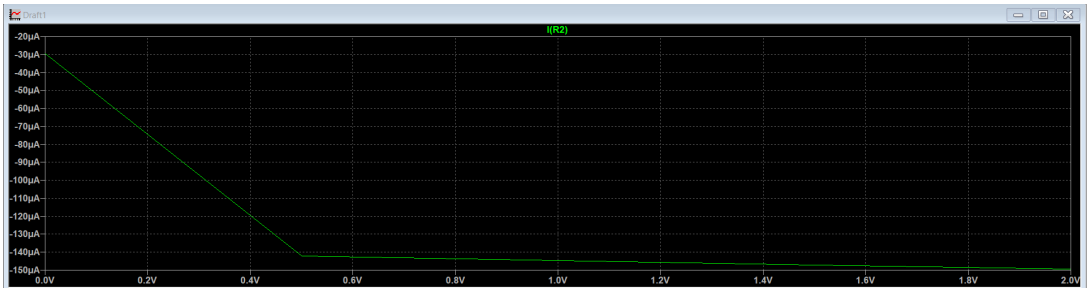


Figure 40: CC Output $I_B = 20\mu A$ (0 to 2V)

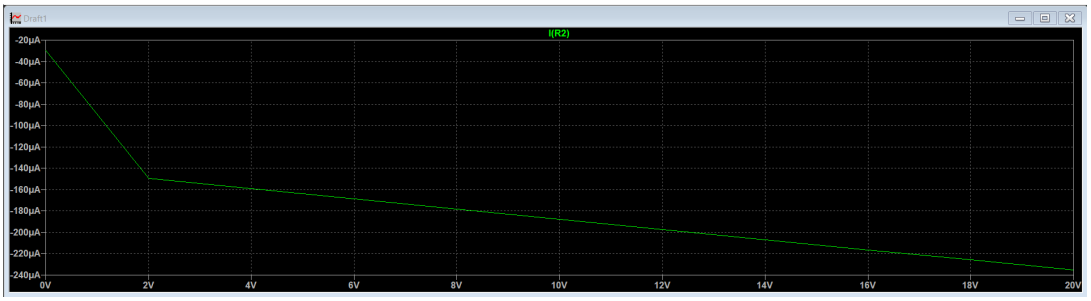


Figure 41: CC Output $I_B = 20\mu A$ (2 to 20V)

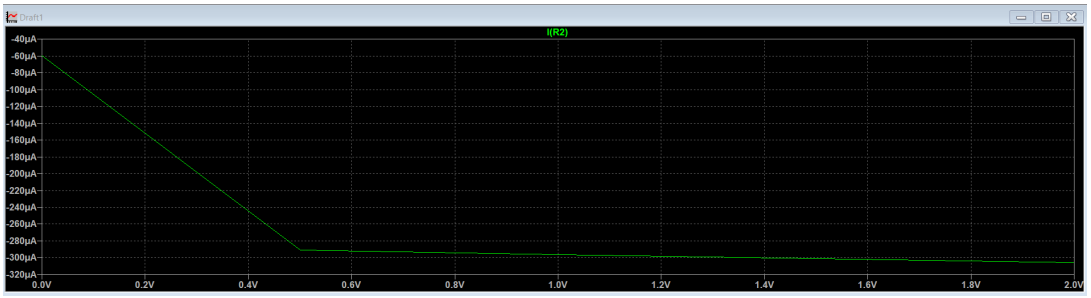


Figure 42: CC Output $I_B = 40\mu A$ (0 to 2V)

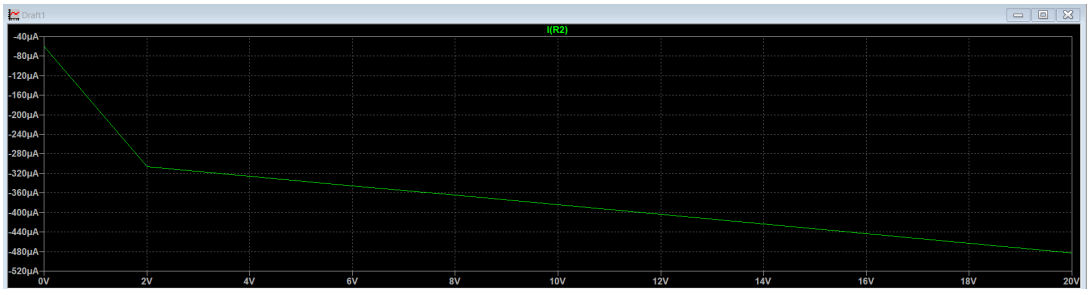


Figure 43: CC Output $I_B = 40\mu A$ (2 to 20V)

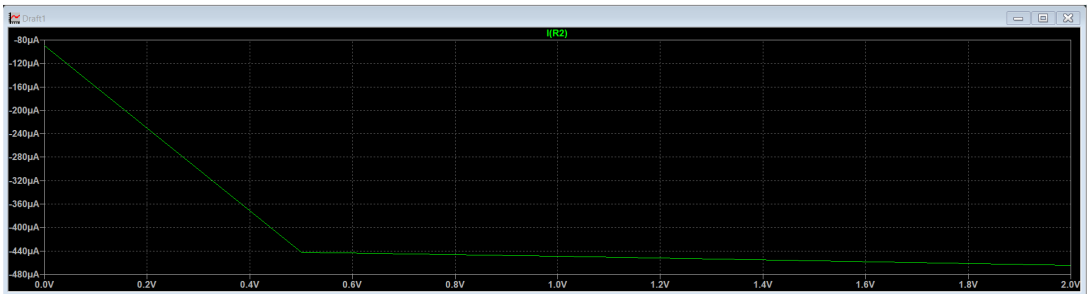


Figure 44: CC Output $I_B = 60\mu A$ (0 to 2V)

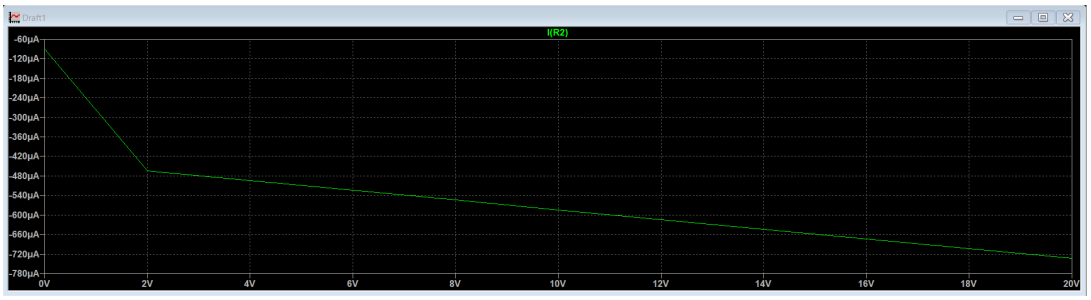


Figure 45: CC Output $I_B = 60\mu A$ (2 to 20V)

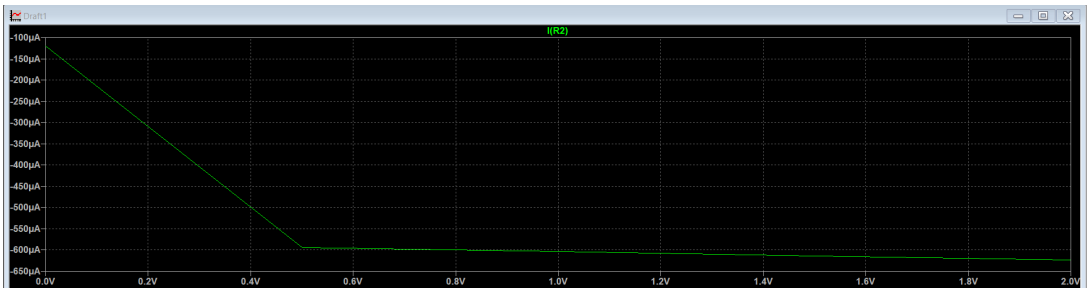


Figure 46: CC Output $I_B = 80\mu A$ (0 to 2V)

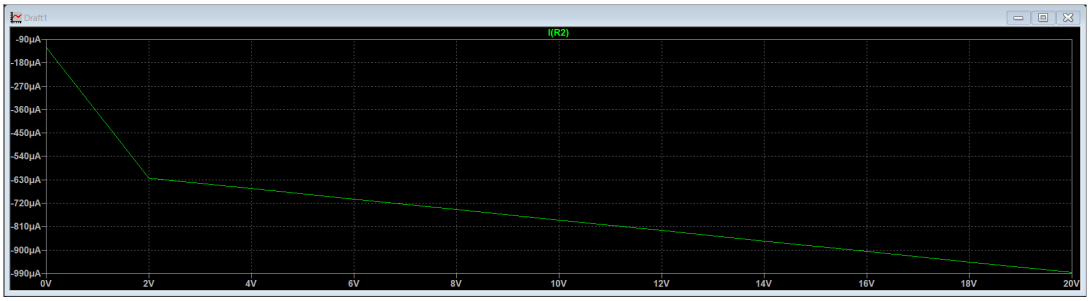


Figure 47: CC Output $I_B = 80\mu A$ (2 to 20V)

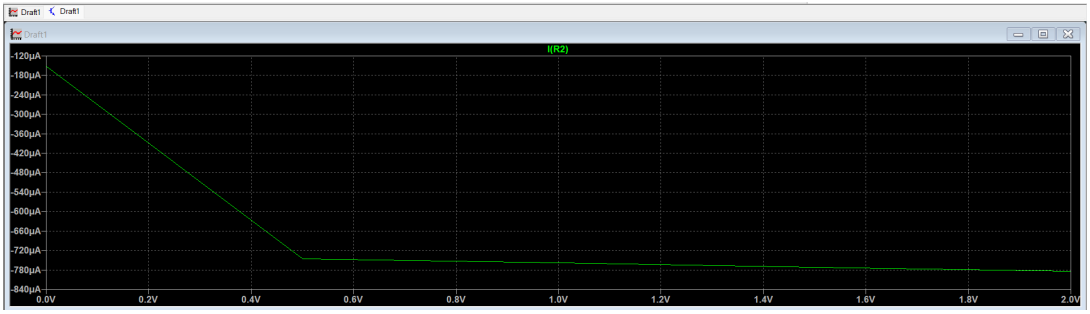


Figure 48: CC Output $I_B = 100\mu A$ (0 to 2V)

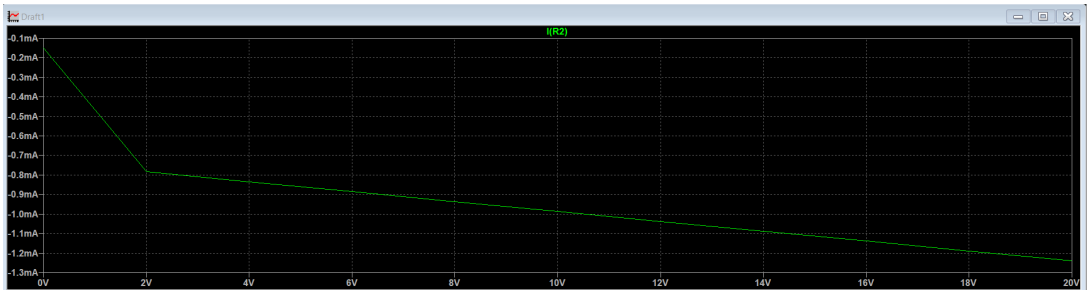


Figure 49: CC Output $I_B = 100\mu A$ (2 to 20V)

5.4 Observation Tables

Table 6: CC Output Characteristics

S.No.	V_{EC} **(V)**	I_E **(mA) at** $I_B = 20\mu A$	I_E **(mA) at** $I_B = 60\mu A$	I_E **(mA) at** $I_B = 100\mu A$
1	0.0	0.0001	0.0001	0.0001
2	0.5	0.0653	0.2245	0.4187
3	1.0	0.0680	0.2326	0.4260
4	1.5	0.0704	0.2396	0.4512
5	2.0	0.0725	0.2463	0.4631
6	4.0	0.0820	0.2743	0.4867
7	6.0	0.0960	0.3261	0.6158
8	8.0	0.4990	0.5947	1324.0

Table 7: CC Input Characteristics

S.No.	V_{BC} **(V)**	I_B **(**μ**A) at** $V_{EC} = 1V$	I_B **(**μ**A) at** $V_{EC} = 5V$	I_B **(**μ**A) at** $V_{EC} = 10V$
1	0.0	0.0000	0.0000	0.0000
2	0.5	0.0014	0.0017	0.0007
3	1.0	0.0023	0.0503	0.0600
4	1.5	0.1041	0.1085	0.1222
5	2.0	0.1642	0.1682	0.1845
6	2.5	0.2234	0.2285	0.2462
7	3.0	0.2825	0.2888	0.3076
8	3.5	0.3426	0.3500	0.3694

6 Calculations / Analysis

Based on the collected data, the following analysis will be performed:

- **CE:** compute $\beta = \Delta I_C/\Delta I_B$ from the output family (at fixed V_{CE}); compute input resistance $r_i = \Delta V_{BE}/\Delta I_B$ and output resistance $r_o = \Delta V_{CE}/\Delta I_C$ from the respective characteristics.
- **CB:** compute $\alpha = \Delta I_C/\Delta I_E$ from the output family (at fixed V_{CB}); compute $r_i = \Delta V_{EB}/\Delta I_E$ and $r_o = \Delta V_{CB}/\Delta I_C$.
- **CC:** compute $\gamma = \Delta I_E/\Delta I_B$ from the output family (at fixed V_{EC}); compute $r_i = \Delta V_{BC}/\Delta I_B$ and $r_o = \Delta V_{EC}/\Delta I_E$.

Cross-check the three current gains using $\beta = \alpha/(1-\alpha)$ and $\gamma = \beta+1$, and compare with the directly measured values. Comment on the relative magnitudes of r_i and r_o across the three configurations (CE: moderate r_i, moderate r_o; CB: low r_i, very high r_o; CC: high r_i, low r_o).

7 Results and Conclusion

The input and output characteristics of the BC547 BJT were simulated in LTSpice for the CE, CB, and CC configurations. The CE configuration demonstrated a moderate slope in the active region indicating a finite output resistance, while the CB configuration showed extremely flat curves representing a high output resistance. Experimental values from the lab will be populated in the observation tables to accurately calculate the practical α, β, and γ values, which will then be verified against their theoretical relationships.